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Integration of Deep Azimuthal Resistivity, Advanced Inversion, and Wired Pipe Technology for Accurate Well Placement in Complex Clastic Reservoirs

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Abstract

Drilling in clastic reservoirs in Middle East presents significant challenges, including variable formation dips, fluctuating layer thicknesses, and complex geological structures. This study demonstrates the integration of Deep Azimuthal Resistivity, Ultra-High Definition (UHD) Inversion Technology, and Wired Pipe Technology to enhance well placement, mitigate drilling risks, and optimize reservoir management. UHD Inversion Technology processes multi-frequency resistivity measurements in real time, producing high-resolution subsurface images. This advanced imaging enhances geological boundary detection, improves structural interpretation, and enables proactive trajectory adjustments, ensuring optimal wellbore placement in complex formations.

The operation targeted a structurally complex clastic reservoir with significant variability. Deep Azimuthal Resistivity provided continuous resistivity measurements, enhancing geological boundary detection and wellbore navigation. UHD Inversion Technology processed real-time resistivity data into high-definition subsurface images, allowing continuous reservoir mapping and precise trajectory adjustments. Wired Pipe Technology enabled high-speed data transmission, ensuring seamless integration of resistivity and inversion data for immediate decision-making. This combination of technologies allowed for dynamic geosteering, reducing uncertainty, and improving operational efficiency in challenging formations.

The integration of these advanced technologies significantly improved well placement and operational efficiency: Deep Azimuthal Resistivity enabled precise detection of geological boundaries, allowing for accurate trajectory corrections and optimal reservoir contact. This minimized the risk of exiting the target zone and enhanced reservoir exposure. UHD Inversion Technology provided continuous, high-resolution subsurface imaging, revealing intricate formation details and structural variations. This allowed for real-time adjustments, ensuring the well remained within the most productive intervals. Wired Pipe Technology facilitated rapid data transmission, eliminating delays associated with conventional telemetry. This ensured real-time collaboration between subsurface experts and drilling teams, leading to faster, more informed geosteering decisions. By leveraging these technologies, the well achieved 100% net-to-gross ratio, maximizing hydrocarbon recovery while minimizing drilling risks.

This case study highlights the effective use of Deep Azimuthal Resistivity, UHD Inversion Technology, and Wired Pipe Technology for precise well placement in clastic reservoirs. The seamless real-time data flow

provided by these technologies set a new benchmark for geosteering precision and operational efficiency in complex geological environments.

Introduction

Drilling operations in complex clastic reservoirs present significant technical challenges due to their inherent geological heterogeneity, structural intricacies, and dynamic depositional histories. Such reservoirs often exhibit variable formation dips, fluctuating layer thicknesses, and complicated geological structures, necessitating advanced drilling methodologies to achieve optimal hydrocarbon recovery. Traditional geosteering methods, typically reliant on standard Logging While Drilling (LWD) tools and conventional mud-pulse telemetry, often fall short in providing real-time precision required to effectively navigate and accurately position wells within such challenging geological environments.

Clastic reservoirs are one of the primary target in Middle East, exemplifies these complexities with its diverse depositional environments ranging from glacially influenced valleys to fluvial, deltaic, and marine settings (Al-Masrahy, 2020; 2024). Characterized by significant variations in reservoir thickness, and complex structural dips ranging from 5° to 10°, therefore, it demands precise wellbore placement strategies to ensure optimal reservoir contact and hydrocarbon recovery (Vianna, 2025; Alshammery, 2017).

To address these challenges effectively, integration of advanced technologies is essential. This paper focuses on the synergistic application of three cutting-edge technologies: Deep Azimuthal Resistivity (DAR), Ultra-High Definition (UHD) Inversion Technology, and Wired Pipe Technology, employed in the drilling operations of case study Well. These technologies collectively provide real-time, high-resolution subsurface imaging, advanced boundary detection capabilities, and immediate data transmission, enabling precise geosteering decisions.

Deep Azimuthal Resistivity (DAR) tools have significantly enhanced the capability to map resistivity contrasts around the wellbore. Utilizing orthogonal antenna configurations, DAR technology delivers robust directional resistivity measurements that enable detection of formation boundaries up to several feet away from the borehole. This facilitates accurate identification and avoidance of undesirable non-reservoir layers, thereby enhancing trajectory adjustments in real time and improving net-to-gross reservoir exposure (Al-Ansari et al., 2024).

Complementing DAR, Ultra-High Definition (UHD) Inversion technology provides unprecedented clarity and resolution in subsurface imaging. UHD inversion processes multi-frequency resistivity measurements through advanced computational algorithms, delivering high-definition images that delineate subtle geological features and thin beds with exceptional precision. This capability significantly reduces geological uncertainty, allowing for proactive, rather than reactive, drilling adjustments (Vianna, 2025).

The introduction of Wired Pipe Technology further revolutionizes the drilling process by overcoming the limitations of conventional mud-pulse telemetry. With data transmission rates exceeding 50,000 bits per second (bps)—far surpassing traditional telemetry methods (Allen et al., 2009)—wired pipe systems enable immediate, high-volume data transmission from downhole to surface. This continuous, high-speed data flow allows drilling teams to collaborate in real time, greatly improving decision-making speed and accuracy, particularly critical in complex geological settings where timely interventions can significantly affect outcomes.

By integrating DAR, UHD Inversion, and Wired Pipe technologies, drilling operations in the complex clastic reservoir

Formation can effectively manage geological uncertainties, optimize well placement accuracy, and enhance overall operational efficiency. This approach not only mitigates drilling risks but also maximizes reservoir contact, ensuring significant improvements in hydrocarbon recovery.

The subsequent sections of this paper will explore the geological background, operational challenges, detailed technology overviews, pre-well planning and feasibility studies, real-time monitoring and

operational results, post-well analysis, and comprehensive conclusions derived from the case study Well. This integration represents a substantial advancement in drilling practices, setting new benchmarks for operational precision and efficiency in complex reservoirs.

Geological Background

The geological complexity of this reservoir is further compounded by significant structural variations and extensive erosional unconformities. Stratigraphically, the reservoir is composed of multiple key layers, each with unique depositional characteristics. A lower layer predominantly consists of glaciogenic sediments, which are overlain by a layer of widespread fluvial and deltaic deposits featuring interbedded sandstones, siltstones, and shales. (Al-Masrahy, 2020, 2024).

Comprehensive geological modeling, supported by 3D seismic data and well log correlations, has been crucial in delineating reservoir facies and optimizing well placement strategies. Accurate mapping of these complex geological features using advanced seismic attributes and resistivity inversion techniques significantly enhances reservoir quality prediction and distribution (Vianna, 2025). The geological complexities of the targeted formation underscore the necessity of integrating advanced geosteering technologies to navigate and exploit these reservoirs effectively.

Challenges and Objectives of the Well

Case study Well encountered numerous geological and operational challenges from variable thickness, which posed a significant challenge in maintaining optimal wellbore positioning within the productive zones, making precision steering crucial.

The zone of interest structure exhibited complex dips varying between 5° to 10°, requiring meticulous trajectory planning and real-time monitoring to avoid deviation from the target zones. Additionally, structural uncertainty due to varying lithological properties and the proximity of resistive, low-porosity roof layers further complicated drilling operations.

Key objectives for case study Well included:

- Maintaining the lateral well trajectory precisely within the targeted interval.
- Maximizing exposure to high-quality sandstones while avoiding penetration of adjacent resistive layers or underlying non-productive zones.
- Implementing continuous real-time monitoring and rapid response adjustments to trajectory based on geological and petrophysical data.

These challenges and objectives underscore the necessity of employing advanced technologies to achieve successful drilling outcomes.

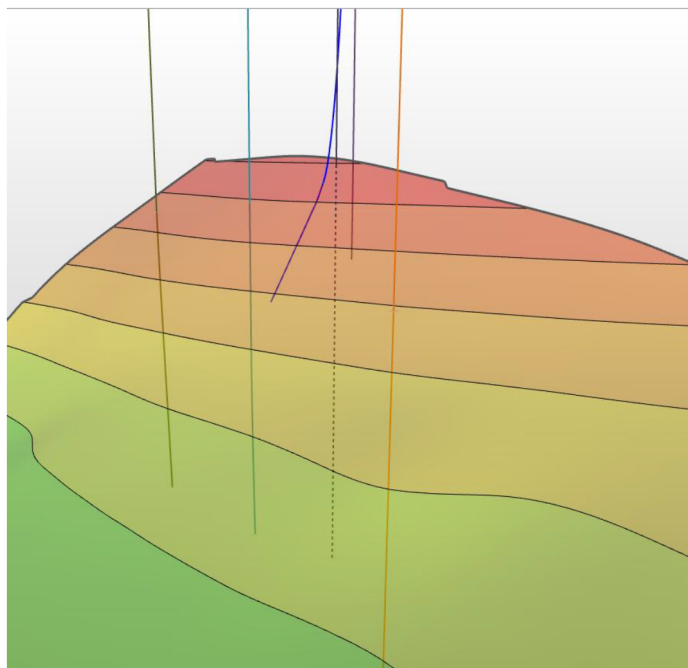


Figure 1—Structural dip and planned trajectory in blue color; map derived from pre-well modeling for the Well.

Technology Overview

Deep Azimuthal Resistivity (DAR)

Deep Azimuthal Resistivity (DAR) technology significantly enhances geosteering capabilities by providing detailed directional electromagnetic measurements. Unlike traditional resistivity tools, which primarily detect formations immediately adjacent to the borehole, DAR utilizes orthogonal and tilted antenna arrays. These antenna arrays enable the detection of resistivity contrasts and geological boundaries at considerable distances from the borehole, often several meters away.

DAR's advanced directional sensitivity allows for real-time detection of critical geological features, including reservoir tops, bottoms, and lateral boundaries. This early detection capability improves trajectory control and reduces geological uncertainty, leading to optimized well placement and higher reservoir contact.

Applications and Advantages

- **Enhanced Well Placement:** Real-time distance-to-bed and apparent-dip measurements significantly improve lateral well positioning, particularly in horizontal and highly deviated wells.
- **Reduction of Non-Productive Time (NPT):** Early boundary detection helps avoid drilling into undesirable zones, minimizing the need for corrective actions such as sidetracks.
- **Improved Reservoir Understanding:** Detailed directional data enhances geological interpretations and reservoir characterization, leading to informed decision-making during drilling operations.

Operational Principles

DAR tools use multiple antenna configurations (orthogonal, coplanar, tilted) that collect electromagnetic data azimuthally around the borehole. As the tool rotates, it continuously measures electromagnetic signals from surrounding formations, allowing operators to determine the direction and distance to resistivity contrasts and formation boundaries.

Technical Specifications

- **Tool Sizes:** Commonly range from 4¾-in. to 6¾-in.
- **Maximum Temperature:** Approximately 150°C (302°F)
- **Maximum Pressure:** Up to 30,000 psi (2070 bar)
- **Detection Depth:** Typically up to 33 feet
- **Resistivity Range:** Capable of measuring from 0.1 to 3,000 ohm-m

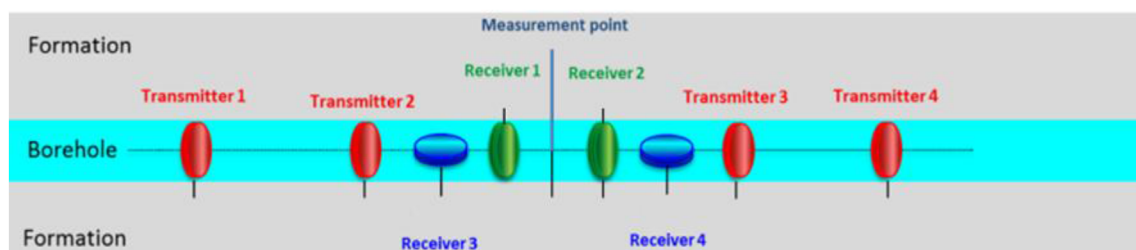


Figure 2—Schematic diagram of typical DAR tool illustrating antenna configurations and directional resistivity measurement capabilities.

Ultra-High Definition (UHD) Inversion

Ultra-High Definition (UHD) Inversion significantly enhances reservoir understanding by providing detailed subsurface resistivity imaging. Unlike standard resistivity inversion, UHD employs multi-frequency, multi-component inversion algorithms, offering significantly improved resolution and sensitivity to subtle geological variations (Vianna, 2025). This technique effectively addresses limitations of conventional inversion methods by implementing dynamic regularization steps to balance resolution and stability, enhancing clarity in identifying thin layers and subtle lithological changes (Vianna, 2025).

Applications and Advantages

- **Enhanced Boundary Detection:** Clearly identifies thin layers and subtle lithological changes.
- **Real-time Geosteering:** Immediate, high-definition feedback facilitates trajectory adjustments during drilling.
- **Geological Uncertainty Reduction:** Enables precise interpretation of complex geological structures.

Operational Principles

UHD inversion processes resistivity measurements from various orientations and frequencies, generating high-resolution, accurate models of the subsurface environment. The dynamic regularization utilized in this method ensures stable, reliable inversion outputs even in complex geological settings, enabling precise geosteering and formation evaluation (Vianna, 2025).

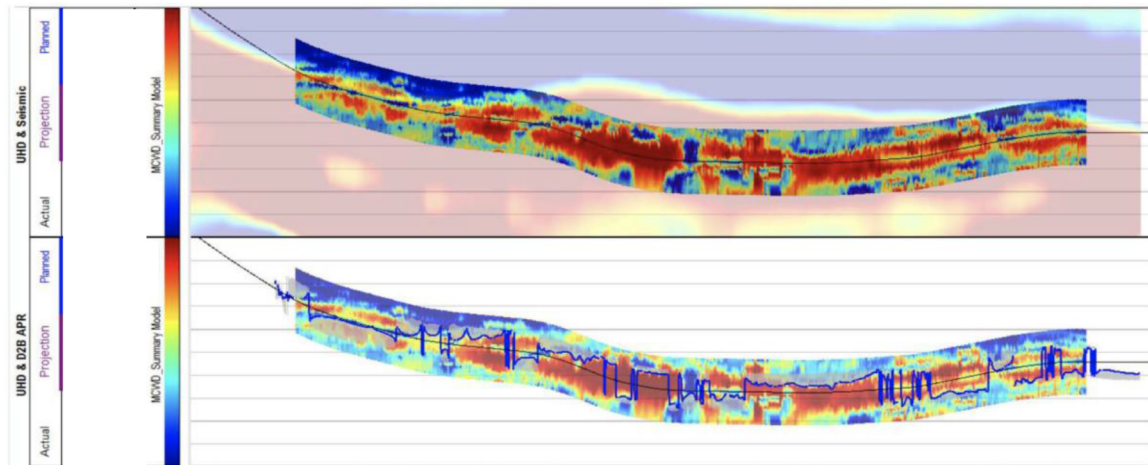


Figure 3—Example of Ultra-High Definition (UHD) inversion output demonstrating enhanced resolution and detailed subsurface resistivity imaging capabilities (Vianna, 2025).

Wired Pipe Technology

Wired Pipe Technology enables high-speed, real-time data transmission through embedded data cables integrated within drill pipes. This technology drastically surpasses conventional mud-pulse telemetry, offering transmission rates up to 57,600 bits per second, greatly reducing data latency and improving reliability under challenging drilling conditions (Allen et al., 2009).

Applications and Advantages:

- Real-time, continuous monitoring of critical drilling parameters.
- Immediate detection and mitigation of drilling hazards, reducing risks significantly.
- Enhanced operational efficiency and reduced non-productive time (NPT) by enabling quicker and informed decisions.

Operational Principles: Wired pipes contain internal data cables and specialized connectors, creating a robust, continuous communication channel from downhole sensors directly to surface systems. This infrastructure ensures immediate and highly reliable data transfer, critical for real-time drilling and formation evaluation operations (Allen et al., 2009).

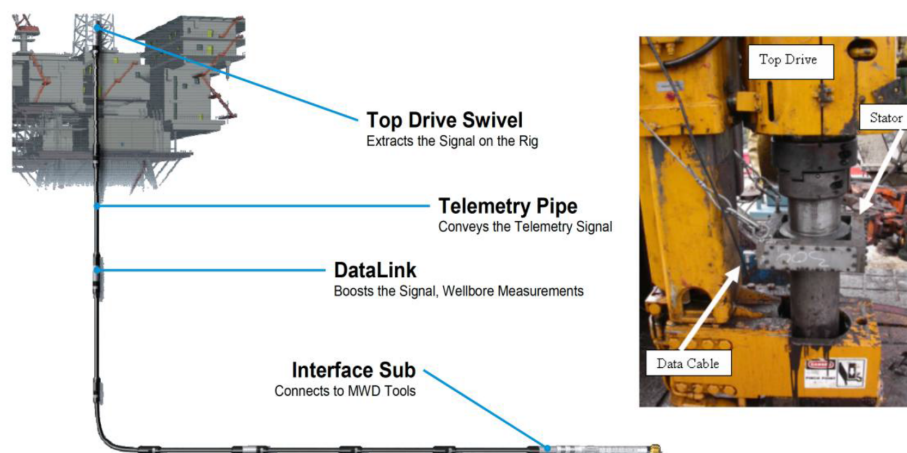


Figure 4—Schematic illustration of wired pipe system configuration, highlighting the internal data cables and communication pathways within the drill string (Allen et al., 2009).

Pre-Well Planning and Feasibility Study

Detailed pre-well planning for case study well was integral to addressing the complex geological conditions anticipated within the target reservoir. The original well plan aimed to drill the 5.875" lateral section towards the southwest, specifically targeting the productive zones. Initial geological and petrophysical models utilized data derived from offset wells, to predict reservoir conditions and optimize trajectory design, (Figure 6).

Pre-well modeling indicated that the reservoir thickness varied significantly. The reservoir was overlain by a dense, highly resistive layer posing a significant challenge to precise well placement. Formation dip trends observed in regional seismic interpretations showed that the structure was consistently dipping downward along the planned well trajectory towards the target Total Depth (TD).

Advanced resistivity inversion modeling predicted that the distinct resistivity contrast at the reservoir's roof would allow for a depth of detection ranging from approximately 7 to 10 feet TVD, providing a critical window for geosteering decisions. This high resistivity contrast, identified through detailed multi-frequency resistivity modeling, provided essential data for pre-planning inclination adjustments and potential steering corrections during real-time operations.

These detailed planning activities, integrating seismic, petrophysical, and geological data, were instrumental in establishing operational strategies designed to navigate geological uncertainties and ensure precise lateral placement within the targeted reservoir interval.

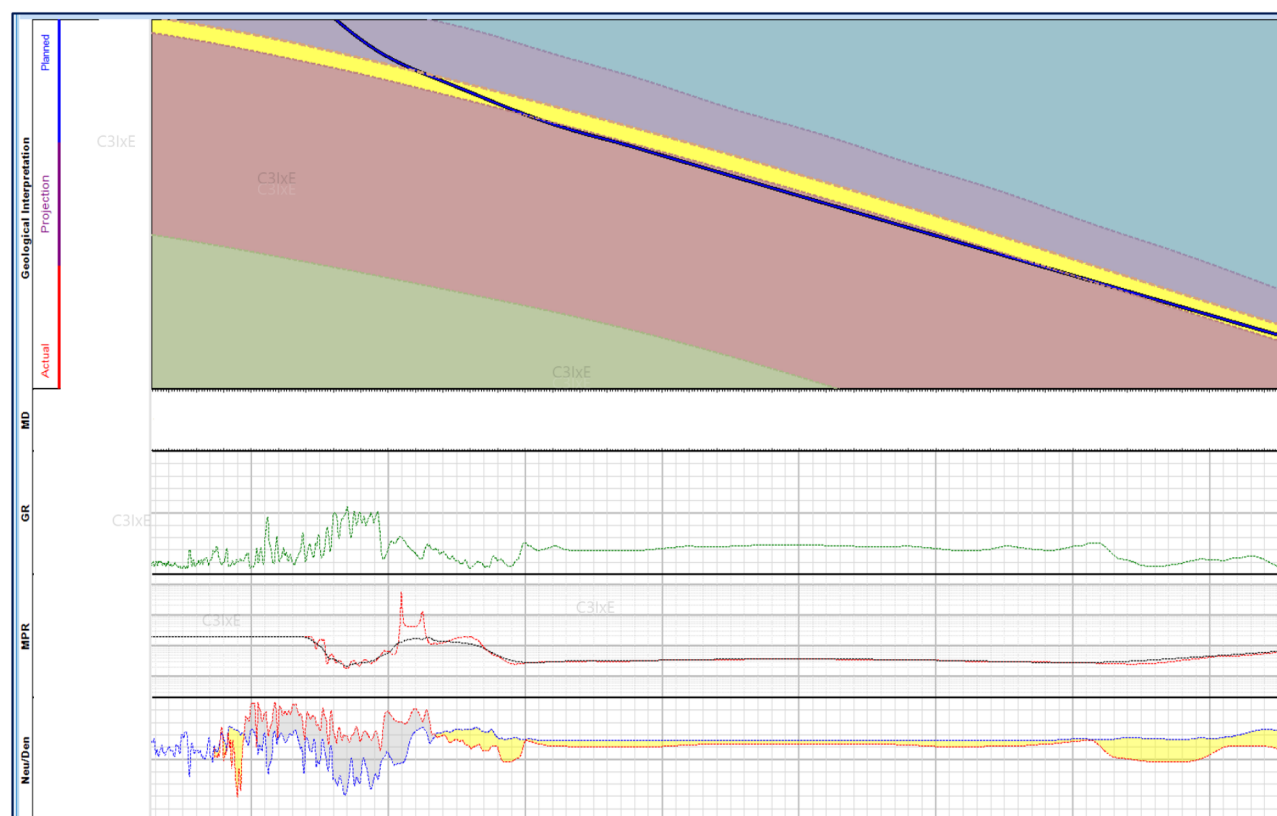


Figure 5—Pre-well modeling cross-section illustrating expected resistivity contrasts and anticipated geological structures derived from offset well data and inversion modeling.

Real-Time Monitoring and Operational Results

Real-time monitoring and operational adjustments were critical to the successful execution of case study Well. Deep drilling operations targeting the zone of interest. Real-time geosteering and continuous data interpretation allowed the team to promptly respond to the complex geological conditions encountered.

Initial monitoring identified formation dip angles varying significantly along the well trajectory, typically ranging between 5° and 10° . Key operational adjustments included strategic inclination changes implemented at critical depths, to maintain the optimal position within the reservoir and avoid the highly resistive roof layer.

Advanced UHD inversion and DAR technologies provided immediate feedback on subsurface geological conditions, facilitating rapid decision-making. These real-time insights were crucial to maintaining trajectory within the high-quality reservoir sands, significantly reducing the risks associated with exiting the productive zone.

Drilling concluded with the wellbore successfully navigating horizontally, precisely placing a significant portion within the targeted interval. The application of advanced real-time monitoring and geosteering technologies notably enhanced operational efficiency, minimized non-productive time, and maximized reservoir contact.

The comparison illustrated in Figure 10 clearly demonstrates the significant enhancement in subsurface imaging achieved through UHD inversion technology. Additionally, the integration of wired pipe technology played a crucial role in increasing the quantity and quality of downhole data points available for inversion processing, directly improving the resolution and accuracy of the UHD inversion model. This allowed for more precise geosteering decisions and optimized well placement within the targeted reservoir interval.

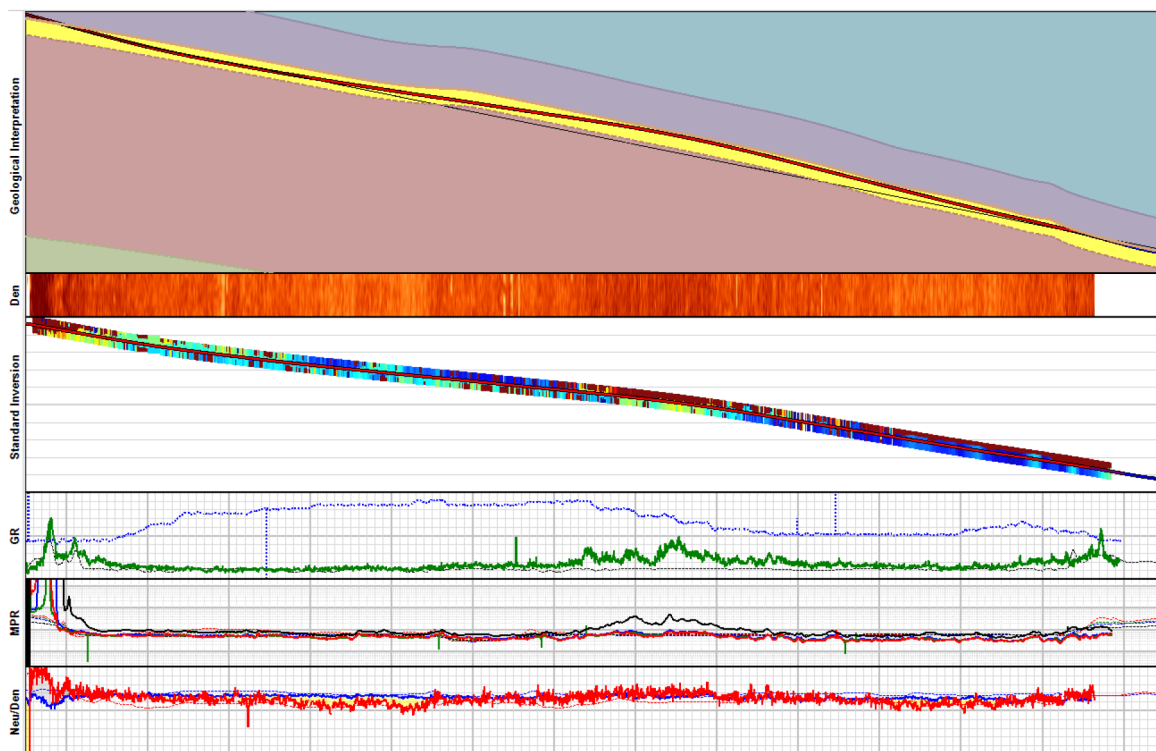


Figure 6—Real-time inversion results and operational trajectory adjustments illustrating precise well placement within the targeted reservoir zone.

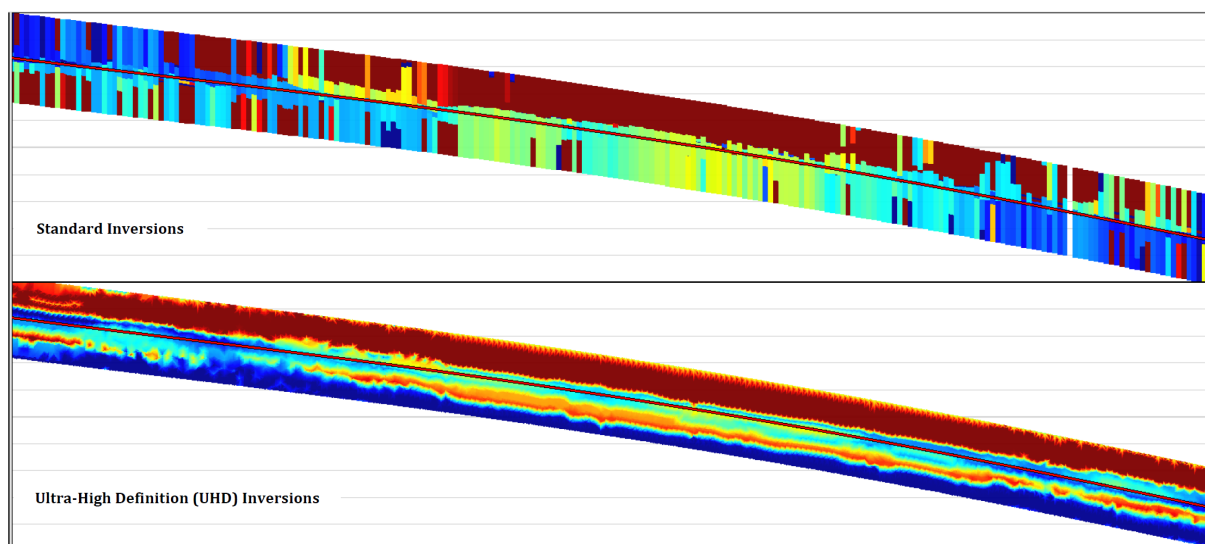


Figure 7—Comparison of standard inversion (up) and Ultra-High Definition (UHD) inversion (down) outputs.

Post-Well Analysis and Discussion

The post-well analysis of case study Well provided valuable insights into the effectiveness and benefits of the integrated technology approach. Detailed analysis confirmed the efficiency of drilling operations and the exceptional accuracy of trajectory adherence within the targeted reservoir interval. Of the horizontally drilled footage, 87% successfully remained within the high-quality productive interval, exceeding initial expectations.

The use of advanced DAR and UHD inversion technologies significantly improved the precision of subsurface imaging and formation boundary detection, enabling proactive, informed geosteering decisions. The integration of wired pipe technology proved instrumental in enhancing the real-time data transmission rate and volume, thereby improving the reliability and accuracy of the UHD inversion process.

Key operational insights from this well included the recognition of the critical role of continuous real-time monitoring to manage geological uncertainty and structural complexity effectively. Immediate feedback enabled prompt, well-informed adjustments to drilling parameters, reducing non-productive time and ensuring maximum reservoir exposure.

Lessons learned underscored the importance of pre-well modeling and real-time adaptive strategies to maintain well trajectory within targeted intervals. The demonstrated operational efficiency and enhanced reservoir contact achieved through this integrated approach provide a robust framework for future drilling operations in similarly complex reservoirs.

Overall, the integration of DAR, UHD inversion, and wired pipe technologies marked a significant advancement in drilling practice, establishing new benchmarks for operational precision, risk mitigation, and reservoir management effectiveness.

Conclusions

The successful drilling and precise well placement of the case study Well highlight the substantial benefits derived from the integrated application of Deep Azimuthal Resistivity (DAR), Ultra-High Definition (UHD) inversion, and Wired Pipe technologies. These advanced technologies collectively contributed to significantly improved operational efficiency, well trajectory accuracy, and optimized reservoir exposure.

The integration of DAR technology enhanced the ability to proactively identify formation boundaries, thereby improving real-time decision-making and trajectory adjustments. The implementation of UHD inversion technology provided superior subsurface imaging, substantially increasing the resolution and continuity mapping of reservoir layers compared to standard inversion methods. The high data transmission

rates achieved through wired pipe technology were critical in maximizing the quantity and quality of downhole data points, thus enhancing the effectiveness and accuracy of the UHD inversion.

Overall, the integrated technological approach established new operational benchmarks, demonstrating remarkable improvements in geosteering precision, risk mitigation, and reservoir management. The successful deployment of these technologies in Well provides valuable insights and a robust framework for future applications in similarly complex geological environments.

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